Answer Key

Physics			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
1	(2)	Vernier Calliper	Units and Measurements
2	(1)	Gauss Law	Electrostatics
3	(3)	Potentiometer	Current Electricity
4	(1)	YDSE, Refraction	Wave Optics
5	(2)	Stefan's Law, Blackbody Radiation	Thermal Properties of Matter
6	(2)	Change of State, Specific Heat, Latent Heat	Thermal Properties of Matter
7	(1)	de Broglie Wavelength	Atoms
8	(2)	Focal Length of Lens, Power of Lens	Ray Optics
9	(3)	Force on Electron in Electric Field	Electrostatics
10	(1)	Resistivity	Current Electricity
11	(3)	Hormonics, Organ Pipes	Waves
12	(4)	Moment of Inertia	System of Particles and Rotational Motion
13	(3)	Gravitational Force	Gravitation
14	(2)	Photoelectric Effect	Dual Nature of Radiations and Matter
15	(2)	Dimensions	Units and Measurements
16	(1)	Conservation of Energy, Verticle Motion	Work, Energy and Power
17	(4)	Combination of Cells in Series and Parallel	Current Electricity
18	(3)	Forward Bias	Semiconductor Electronics
19	(3)	Parallel Plate Capacitor	Electrostatics
20	(4)	Focal Length of Lens, Power of Lens	Ray Optics
21	[4]	Excess Pressure	Mechanical Properties of Fluids
22	[8]	Mirror Formula, Image Formation by Mirror	Ray Optics
23	[40]	Thermal Conductivity	Thermal Properties of Matter
24	[90]	Angular Momentum, Vectors	Vectors
25	[5]	Equations of Motion	Motion in 1-D

Chemistry			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
26	(3)	Faraday Law of Electrolysis	Electrochemistry
27	(3)	First Order Kinetics	Chemical Kinetics
28	(4)	Stereoisomerism	Isomerism
29	(1)	Electronegativity	Periodic Classification of Elements
30	(1)	Electronic Configuration	d and f Block Elements
31	(3)	Periodic Trends	Periodic Classification of Elements
32	(3)	Vitamin	Biomolecules
33	(1)	General Terms of Thermodynamics	Thermodynamics
34	(4)	Bohr Radius	Structure of Atom
35	(4)	Nucleophilic Substitution Reaction	Haloalkane and Haloarene
36	(1)	Acidic Nature of Alkyne	Hydrocarbon

37	(2)	Calculation of Equilibrium Constant	Chemical Equilibrium
38	(4)	IUPAC	Nomenclature
39	(3)	Product of Electrolysis	Electrochemistry
40	(3)	Fehling's Test	Practical Organic Chemistry
41	(4)	CFSE Calculation	Coordination Chemistry
42	(1)	Elevation in Boiling Point	Liquid Solutions
43	(3)	Properties of Diazonium Salts	Nitrogen Containing Compounds
44	(2)	Geometry and Oxidation State of Complex	Coordination Chemistry
45	(1)	Properties of Alkene	Hydrocarbon
46	[45]	Acid Base Titration	Ionic Equilibrium
47	[20]	Carius Method	Purification of Organic Chemistry
48	[6]	Geometry of Molecule or Ion	Chemical Bonding and Molecular Structure
49	[69]	Arrhenius Equation	Chemical Kinetics
50	[154]	Properties of Diazonium Salts	Amines and Haloalkane

Mathematics

Q. No.	Answer	Topic Name	Chapter Name
51	(3)	Equivalence Relation	Relation and Function
52	(1)	Functional Rule	Function
53	(4)	Point and Line	Straight Line
54	(4)	Algebra of Complex Number	Complex Number
55	(4)	Range	Inverse Trigonometric Function
56	(2)	Probability Distribution	Probability
57	(3)	Geometrical Progression	Sequence and Series
58	(4)	Shortest Distance Between Lines	3D
59	(1)	Logarithmic Equation	Logarithmic Indices
60	(3)	Sum of Special Series	Sequence and Series
61	(4)	Permutations	Permutations and Combination
62	(3)	Linear Differential Equation	Differential Equations
63	(2)	Parabola	Conic Section
64	(1)	Circle and Line	Circle
65	(3)	Properties of Definite Integral	Definite Integral
66	(2)	Functional Equation	Differentiation
67	(1)	Application on Sets	Sets
68	(3)	Area Between Curves	Application of Integral
69	(1)	Bayes Theorem	Probability
70	(3)	Equation of Hyperbola	Hyperbola
71	[34]	Differentiability of a Function	Continuity and Differentiability
72	[2035]	Properties of Binomial Coefficients	Binomial Theorem
73	[34]	Properties of Adjoint	Determinants
74	[216]	Line	Three Dimensional
75	[16]	Cross Product	Vector

Answers with Explanation

Physics

1. Option (2) is correct.

The general principle of vernier calipers is N (vernier scale divisions) = (N − 1) main scale divisions.

i.e., N(VSD) = (N−1) MSD

$$\text{MSD} = \left(\frac{N}{N-1}\right)\text{VSD}$$

since, $N > (N-1)$

$\therefore$ 1 MSD > 1 VSD

Statement I is true.

Vernier constant (VC) = 1 MSD − 1 VSD

Statement II is false.

2. Option (1) is correct.

When a charge is placed on center of an edge of a cube, electric flux comes out of the other four faces, which do not share this charge. In the given question, the charge placed along the edge 'BC' is shared by two faces of the cube, 'ABCD' and 'BCHG.'

Moreover, to enclose this charge, we need to imagine four cubes. Hence, flux per face out of sixteen faces is $= \frac{q}{16\varepsilon_0}$.

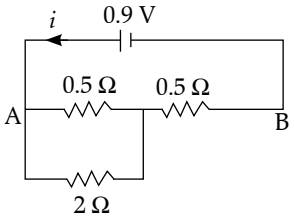
Flux coming out of the four faces of single cube is $4\times\frac{q}{16\varepsilon_0}$. Using $q = \lambda\left(\frac{a}{2}\right)$, we get

$$\phi_E = \frac{4}{16\varepsilon_0}\times\frac{\lambda a}{2} = \frac{\lambda a}{8\varepsilon_0}.$$

3. Option (3) is correct.

The circuit can be redrawn as:

$$R_{AB} = \frac{2\times0.5}{2.5} + 0.5$$
$$= 0.4 + 0.5$$
$$= 0.9\ \Omega$$
$$i = \frac{V}{R} = \frac{0.9}{0.9} = 1\text{A}$$



4. Option (1) is correct.

Fringe width is given by,

$$\beta = \frac{\lambda D}{d}$$

In an optically dense medium, the wavelength decreases i.e. $\lambda' < \lambda$. As $\beta \propto \lambda$ fringe width will also decrease with decrease in wavelength

In air, $c = f\lambda$.

In a medium $v = f\lambda$, as frequency does not change with the medium. Hence, $v \propto \lambda$

Since, $\lambda' < \lambda$ the reason is also true and completely explains the assertion.

5. Option (2) is correct.

Power (energy per unit time) $= \sigma e T^4 \times 4\pi R^2$,

where σ is the Stefan's constant.

$$\frac{E}{E'} = \left(\frac{T_1}{T_2}\right)^4\left(\frac{R_1}{R_2}\right)^2 = \left(\frac{800}{400}\right)^4\left(\frac{0.2}{0.8}\right)^2 = 16\times\frac{1}{16} = 1$$

$E' = E$

6. Option (2) is correct.

w_1 = work done to raise temp to 0°C

$= ms_{ice}\,\Delta T = 0.001 \times 2100 \times (0 + 10) = 21$ J

$w_2 = mL_{ice} = 0.001 \times 3.35 \times 10^5 = 335$ J

$w_3 = ms_{water}\,\Delta T = 0.001 \times 4180 \times (100–0 = 418$ J

$w_4 = ms_{stream} = 0.001 \times 2.25 \times 10^6 = 2250$ J

$w_5 = ms_{stream}\,\Delta T = 0.001 \times 1920 \times (110 – 100)$

$=19.2$ J

Total work done, heat $= w_1 + w_2 + w_3 + w_4 + w_5$

$\Rightarrow W = H = 21 + 335 + 418 + 2250 + 19.2$

$= 3043.2$ J ~ 3043 J

7. Option (1) is correct.

According to Bohr's theory, for H atom

$r = r_0 n^2$

where, r_0 = radius in the ground state,

n = orbit number,

$$v = \frac{v_0}{r_1}$$

where, v_0 = velocity in the ground state,

de Broglie wavelength $\lambda = \frac{h}{mv}$

$\frac{\lambda_0}{\lambda_4} = \frac{v_4}{v_0}$ as for the 3rd excited state, $n = 4$

$$\frac{\lambda_0}{\lambda_4}=\frac{v_0/4}{v_0}=\frac{1}{4}$$

Note $\frac{\lambda_4}{\lambda_0}=4$

8. Option (2) is correct.

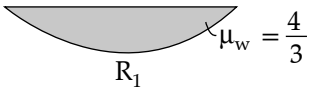
The given arrangement can be considered as combination of three lenses kept in contact. The power of the combination is given by

$$P=\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}$$

1st lens

$$\frac{1}{f_1}=(\mu_w-1)\left(\frac{1}{\infty}-\frac{1}{-R_1}\right)$$

$$=\left(\frac{4}{3}-1\right)\left(\frac{1}{R_1}\right)$$

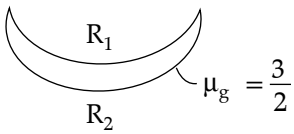


$$\frac{1}{f_1}=\frac{1}{3R_1}$$

2nd lens

$$\frac{1}{f_2}=(\mu_g-1)\left(\frac{1}{-R_1}-\frac{1}{-R_2}\right)$$

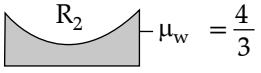
$$=\left(\frac{3}{2}-1\right)\left(\frac{-1}{R_1}+\frac{1}{R_2}\right)$$



$$\frac{1}{f_2}=\frac{1}{2}\left(\frac{-1}{R_1}+\frac{1}{R_2}\right)$$

3rd lens

$$\frac{1}{f_3}=(\mu_w-1)\left(\frac{1}{-R_2}-\frac{1}{\infty}\right)$$



$$=\left(\frac{4}{3}-1\right)\left(-\frac{1}{R_2}\right)$$

$$\frac{1}{f_3}=\frac{-1}{3R_2}$$

By inserting all values, we get:

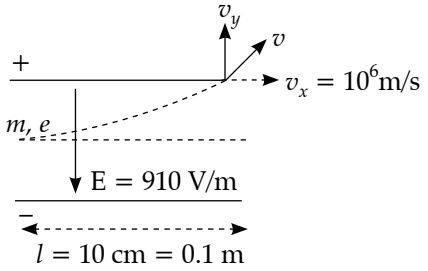
$$P=\frac{1}{3R_1}+\frac{1}{2}\left(\frac{-1}{R_1}+\frac{1}{R_2}\right)-\frac{1}{3R_2}$$

$$=\frac{1}{R_1}\left(\frac{1}{3}-\frac{1}{2}\right)+\frac{1}{R_2}\left(\frac{1}{2}-\frac{1}{3}\right)$$

$$=\frac{-1}{6R_1}+\frac{1}{6R_2}$$

$$=\frac{-1}{6}\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

9. Option (3) is correct.



Since, there is no force acting along the x-axis,

$a_x = 0$

$u = v_x = 10^6$ m/s

Along y direction $a_y=\frac{eE}{m}$

$\therefore v_y = u_y + a_y t$

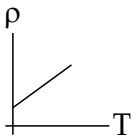
where, $t=\frac{l}{v_n}=\frac{0.1}{10^6}=10^{-7}$ s

$$\therefore \quad v_y=0+\frac{eE}{m}t$$

$$v_y=\frac{1.6\times10^{-19}\times910\times10^{-7}}{9.1\times10^{-31}}=16\times10^6\,\text{m/s}$$

10. Option (1) is correct.

Since, most of the wire-bound resistors are always a straight line made up of alloys



The variation of ρ with T,

$\rho = \rho_0 (1 + \alpha \Delta T)$

where, α temperature is the coefficient of resistance.

11. Option (3) is correct.

The velocity of sound in air $v=\sqrt{\frac{B}{\rho}}$.

The 9th harmonic of the closed tube = the 4th harmonic of the open tube.

$$9\frac{v_c}{4l_c}=4\frac{v_0}{2l_0}$$

$$\Rightarrow \frac{9}{4l_c}\sqrt{\frac{B}{\rho_1}}=\frac{4}{2l_o}\sqrt{\frac{B}{\rho_2}}$$

$$\Rightarrow l_o=\frac{2\times40}{9}\sqrt{\frac{\rho_1}{\rho_2}}=\frac{80}{9}\times\frac{1}{4}=\frac{20}{9}\text{cm}$$

12. Option (4) is correct.

Removed mass $M'=\frac{M}{\pi R^2}\times\pi\left(\frac{R}{2}\right)^2=\frac{M}{4}$

As per parallel axis theorem:

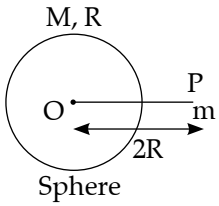
$$I_0=\frac{MR^2}{2}-\left[\frac{M'}{2}\left(\frac{R}{2}\right)^2+M'\left(\frac{R}{2}\right)^2\right]$$

$$= \frac{MR^2}{2} - \frac{3}{8}M'R^2$$

$$= \frac{MR^2}{2} - \frac{3}{32}MR^2$$

$$= \frac{13}{32}MR^2$$

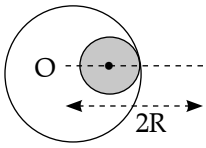
13. Option (3) is correct.



Initially, $F_1 = \frac{GMm}{(2R)^2}$

Removed mass:

$$M' = \frac{M}{\frac{4}{3}\pi R^3} \times \frac{4}{3}\pi\left(\frac{R}{3}\right)^3 = \frac{M}{27}$$



$$F_2 = \frac{GMm}{(2R)^2} - \frac{GM'm}{\left(R + \frac{R}{3}\right)^2}$$

$$= \frac{GMm}{4R^2} - \frac{G\left(\frac{M}{27}\right)m}{\frac{16R^2}{9}}$$

$$= \frac{GMm}{R^2}\left(\frac{1}{4} - \frac{9}{16 \times 27}\right)$$

$$= \frac{GMm}{R^2}\left(\frac{1}{4} - \frac{1}{48}\right)$$

$$F_2 = \frac{11}{48}\frac{GMm}{R^2}$$

$$\frac{F_1}{F_2} = \frac{1}{4} \times \frac{48}{11} = \frac{12}{11}$$

14. Option (2) is correct.

Threshold wavelength $\lambda_0 = \frac{1240}{E(\text{eV})}(\text{nm})$

For Cs, $\lambda_{01} = \frac{1240}{1.9} \approx 653\,\text{nm}$

For Li, $\lambda_{02} = \frac{1240}{2.5} \approx 496\,\text{nm}$

Photoelectric emission take place only when the incident wavelength $\lambda < \lambda_0$.

As, λ_{02} (Li) $< \lambda < \lambda_{01}$ (Cs)

Hence, the photoelectric effect is possible for Cs only.

15. Option (2) is correct.

The magnetic field (B) related to permeability of free space (μ_0) is given by:

$$B = \mu_0 ni$$

$$= \mu_0 \frac{Ni}{l}$$

$$\left[\frac{B}{\mu_0}\right] = \left[\frac{i}{n}\right] = \left[AL^{-1}\right]$$

16. Option (1) is correct.

At point 'D,' T+ $mg = \frac{mv_d^2}{l}$.

Since, the string becomes slack at 'D', T = 0

$v_d^2 = gl$.

COE gives $\frac{1}{2}mv_0^2 = \frac{1}{2}mv_d^2 + mg(2l)$

$$\frac{1}{2}mv_0^2 = \frac{1}{2}m(gl) + mg(2l)$$

$$\frac{1}{2}mv_0^2 = \frac{5}{2}mgl$$

$v_0 = \sqrt{5gl}$

At any θ measured from bottom vertical,

$v^2 = v_0^2 + 2gl\cos\theta - 2gl$

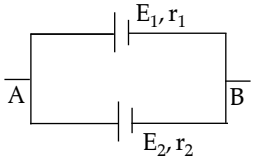
At point 'B,' $\theta = 60°$, $v_B^2 = 5gl - gl = 4gl$

At point 'C' $\theta = 120°$, $v_c^2 = 5gl - 3gl = 2gl$

$$\frac{KE_B}{KE_C} = \frac{v_B^2}{v_C^2} = 2:1$$

17. Option (4) is correct.

$$V_{AB} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}$$



$$r_{AB} = \frac{r_1 r_2}{r_1 + r_2}$$

Here, $E_1 = E$, $E_2 = 2E$, $r_1 = r$, $r_2 = 4r$

$V_{AB} = 1.2E < E_1$

$R_{AB} = \frac{4r}{5} = 0.8r < r_1$ (as well as r_2).

Statement I is false but Statement II is true.

18. Option (3) is correct.

A diode is said to be forward-biased biased if $V_P > V_N$.

As can be seen, this is valid for figures in B, C and E.

19. Option (3) is correct.

On inserting the dielectric slab, the capacitance increases to $C = KC_0 = 2 \times 40 = 80\ \mu F$

Initial change $q_i = C_0V$

Final change $q_f = CV$

Extra charge $= q_f - q_i = (C - C_0)V$

$= (80 - 40) \times 100$

$= 4000\ \mu C = 4\ mC$

$$\Delta U = \frac{1}{2}(C - C_0)V^2 = \frac{1}{2}(80 - 40) \times (100)^2$$

$$= 20 \times 10^4\ \mu J = 0.2\ J$$

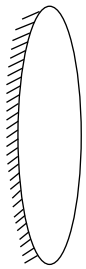
20. Option (4) is correct.

A silvered (polished) lens behaves like a concave mirror the focal length of which is given by:

$$\frac{1}{f} = \frac{-2}{f_e} + \frac{1}{f_m}$$

$$\frac{1}{f_e} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{-R_2}\right) = 2\left(\frac{\mu - 1}{R}\right)$$

$$\frac{1}{f_m} = \frac{2}{-R}$$



$$\therefore \frac{1}{f} = -4\left(\frac{\mu - 1}{R}\right) - \frac{2}{R} = \frac{-4\mu}{R} + \frac{2}{R}$$

$$\frac{1}{f} = \frac{2 - 4\mu}{R}$$

$$f = \frac{R}{2 - 4\mu}(-ve)$$

$$|f| = \frac{R}{4\mu - 2}$$

The image coincides with the object when the object is placed at the centre of the curvature of the equivalent concave mirror.

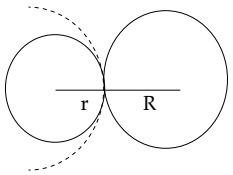
$$R' = 2|f| = \frac{R}{-1 + 2\mu}$$

21. Correct answer is [4].

Due to surface tension, pressure is more on the concave side of the meniscus excess pressure is given by.

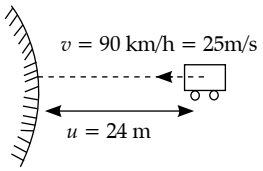
$$\frac{4S}{r} - \frac{4S}{R} = \frac{4S}{r'}$$

where, r' is the radius of curvature of the common surface.



$$r' = \frac{Rr}{R - r} = \frac{4 \times 2}{4 - 2} = 4\text{cm}$$

22. Correct answer is [8].



$R = 2$ m

$$f = \frac{R}{2} = 1\text{ m}$$

Using the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ ...(i)

$$\frac{1}{v} + \frac{1}{-24} = \frac{1}{1}$$

$$v = \frac{24}{25}\text{m}$$

Differentiating eqn (i) on both sides w.r.t. t

$$\frac{-1}{v^2}\frac{dv}{dt} - \frac{1}{u^2}\frac{du}{dt} = 0$$

$$\frac{-1}{v^2}v_i - \frac{1}{u^2}v_0 = 0$$

$$v_i = -\left(\frac{v^2}{u^2}\right)v_0$$

where v_i = velocity of image

v_0 = velocity of object

$$= -\left(\frac{v}{u}\right)^2 v_0$$

$$= -\left(\frac{24/25}{-24}\right)^2 \times (-25)$$

$$v_i = \frac{1}{25}\text{m/s}$$

$$v_i = -\left(\frac{v^2}{u^2}\right)v_0 = -\left(\frac{v}{u}\right)^2 v_0$$

Differentiating again w.r.t. t

$$\frac{dv_i}{dt} = a_i = -v_0\left(2\frac{v}{u}\right)\left[\frac{uv_i - vv_0}{u^2}\right]$$

$$= -\frac{2vv_0}{u^3}\left[uv_i - vv_0\right]$$

By inserting all the values, we get:

$$a_i = \frac{-2\left(\frac{24}{25}\right)(-25)}{(-24)^3}\left[-24\left(\frac{1}{25}\right) - \frac{24}{25}(-25)\right]$$

$$= -\frac{1}{12 \times 24}\left[\frac{-1}{25} + 1\right] \times 24$$

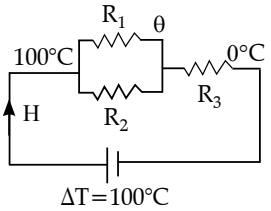
$$= \frac{-1}{12} \times \frac{24}{25}$$

$= \frac{-2}{25}\text{m/s}^2$

$|100a_1| = 100 \times \frac{2}{25} = 8\text{m/s}^2$

23. Correct answer is [40].

The given arrangement can be converted into an equivalent circuit as shown below.



$R_1 = \frac{l}{k_1 A}$

$R_2 = \frac{l}{k_2 A}$

$R_3 = \frac{l}{k_3(2A)}$

Heat current $H = \frac{\Delta T}{R}$

where, $R = \frac{R_1 R_2}{R_1 + R_2} + R_3 = \frac{l}{180A} + \frac{l}{135(2A)}$

And $\theta = HR_3$

$= \frac{100}{\frac{l}{A}\left(\frac{1}{180} + \frac{1}{270}\right)} \times \frac{l}{2A(135)} = \frac{100 \times 180 \times 135 \times 2}{2 \times 135 \times 450}$

$\theta = 40°\text{C}$

24. Correct answer is [90].

Given, $m_A = m_B = 1$ kg

$\vec{r_A} = t^2\hat{i} + 3nt\hat{j} + 2t\hat{k}$

$\vec{r_B} = 2t\hat{i} + t^2\left(-\hat{j}\right) + 4pt\hat{k}$

$\vec{v_A} = \frac{dr_A}{dt} = 2t\hat{i} + 3n\hat{j} + 2\hat{k}$

$\vec{v_B} = \frac{dr_B}{dt} = 2\hat{i} - 2t\hat{j} + 4p\hat{k}$

At $t = 1$ s, $|\vec{v_A}| = |\vec{v_B}|$

$\vec{v}_A = 2\hat{i} + 3n\hat{j} + 2\hat{k}$, $\vec{r}_A = \hat{i} + 3n\hat{j} + 2\hat{k}$

$\vec{v}_B = 2\hat{i} - 2\hat{j} + 4p\hat{k}$, $\vec{r}_B = 2\hat{i} - \hat{j} + 4p\hat{k}$

$\sqrt{4 + 9n^2 + 4} = \sqrt{4 + 4 + 16p^2}$

$9n^2 = 16p^2 \Rightarrow 3n = \pm 4p$ ...(i)

Also, at $t = 1$s,

$\vec{v}_A \cdot \vec{v}_B = 0$

$4 - 6n + 8p = 0$

$8p - 6n = -4$

$3n - 4p = 2$ ...(ii)

Now, from equation (i) $3n = 4p$ makes equation (ii) meaningless.

Hence, $3n = -4p$ gives $p = \frac{-1}{4}$ and $n = \frac{1}{3}$

Now, $\vec{r}_{AB} = \vec{r}_A - \vec{r}_B = -\hat{i} + (3n+1)\hat{j} + (2-4p)\hat{k}$

$= \hat{i} + 2\hat{j} + 3\hat{k}$

Angular momentum $\vec{L}_A = m_A(\vec{r}_{AB} \times \vec{r}_A)$

$$\vec{L}_A = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 3 \\ 2 & 1 & 2 \end{vmatrix}$$

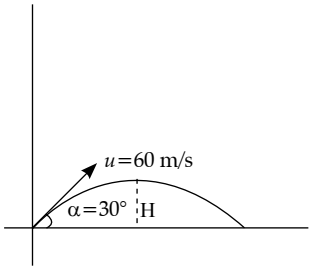
$= \hat{i}(4-3) - \hat{j}(-2-6) + \hat{k}(-1-4)$

$= \hat{i} + 8\hat{j} - 5\hat{k}$

$|\vec{L}_A| = \sqrt{90} = \sqrt{L}$

$\therefore$ L = 90

25. Correct answer is [5].



Maximum height

$H = \frac{u^2 \sin^2\theta}{2g} = \frac{3600 \times \frac{1}{4}}{20} = 45\text{m}$

Time of light $T = \frac{2u\sin\theta}{g} = \frac{2 \times 60 \times \frac{1}{2}}{10} = 6\text{s}$

$\frac{T}{2} = 3s$

Height reached in the first second

$h_0 = (u\sin\theta)(1) - \frac{1}{2}g(1)^2$

$= \left(60 \times \frac{1}{2}\right) - 5$

$= 25$ m

Height reached in the first two seconds

$= (u\sin\theta) \times 2 - \frac{1}{2}g(2)^2 = 40$ m

Distance covered in the last second before it reaches the maximum height $= H - h_2 = h_1$

$45 - 40 = 5$ m

Now, $\frac{h_0}{h_1} = \frac{25}{5} = 5$

Chemistry

26. Option (3) is correct.

Formula used: $w = \frac{Eit}{96,500}$

Where E = Equivalent Weight of Aluminium

$= \frac{\text{Atomic Weight}}{\text{Oxidation State}} = \frac{27}{3} = 9\text{g}$

$t = 30 \text{ min} = 30 \times 60 = 1800 \text{ s}$

$i = 2\text{A}$

$w = \frac{9 \times 2 \times 1800}{96,500} = \frac{18 \times 1800}{96,500}$

$= 0.336 \text{ g}$

27. Option (3) is correct.

Radioactive decay is a first order.

Kinetic reaction, in which the rate of reaction depends on the first power of concentration, i.e., Rate = k (radioactive material)

The integrated rate equation of first order is as follows:

$k = \frac{2.303}{t} \log \frac{a_0}{(a_0 - x)}$

From above, it is clear that the decay constant (k) is independent of the temperature, so the statement decay constant increases with increase in temperature is incorrect.

28. Option (4) is correct.

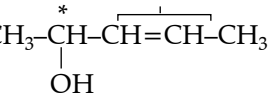
For asymmetrical molecule, the number of different isomers can be calculated as:

No. of stereoisomers $= 2^n$

Where,

n = No. of stereogenic centre

Here, $n = 2$



No. of isomers = $2^2 = 4$ isomers

29. Option (1) is correct.

The tendency to attract shared pair of e^- to words itself in a single covalent bond is defined as electronegativity.

In a periodic table, electronegativity increase across the period while decreases down the group.

(1) B < N 2nd period

Mg < Al 3rd period

Correct order of electronegativity is:

Mg < Al < B < N

(2) C < N 2nd period

↓

Al < Si 3rd period

Correct order of electronegativity is:

Al < Si < C < N

(3) Be <B < N 2nd period

↓

Mg 3rd period

Correct order of electronegativity is

Mg < Be < B < N

(4) O ← F 2nd period

S → Cl 3rd period

Correct order of electronegativity is:

S < Cl < O < F

From the above, it is clear that the incorrect order of electronegativity is:

Al < Mg < B < N

30. Option (1) is correct.

Element	Symbol	Configuration	Ion Configuration
Europium	Eu	$4f^7 6s^2$	$Eu^{2+} = 4f^7$
Gadolinium	Gd	$4f^7 5d^1 6s^2$	$Gd^{3+} = 4f^7$
Europium	Eu	$4f^7 6s^2$	$Eu^{3+} = 4f^6$
Terbium	Tb	$4f^9 6s^2$	$Tb^{3+} = 4f^8$
Samarium	Sm	$4f^6 6s^2$	$Sm^{2+} = 4f^6$

From above, it is clear that the lanthanoid ion with $4f^7$ configuration are Eu^{2+} and Gd^{3+}.

31. Option (3) is correct.

(A) $Al^{3+} < Mg^{2+} < Na^+ < F^-$ (IV) Ionic radii

On increasing positive charge ionic radii decreases

$\therefore$ $A1^{3+}$ having smallest radii.

(B)B < C<O< N (I) Ionisation enthalpy

The ionisation enthalpy of nitrogen action is greater than oxygen atom due to stable half-filled configuration.

(C) B < Al < Mg < K (II) Metallic character

The metallic character increases top to bottom in the group, while decreases left to right across the period.

(D) Si < P < S < Cl (III) Electronegativity

Si P S Cl 3rd period.

The electronegativity value increases across the period as the atomic radii decreases.

32. Option (3) is correct.

Vitamin C is also known as ascorbic acid. It is water soluble vitamin, which must be supplied regularly in diet as they are readily excreted in urine and cannot be stored in our body.

33. Option (1) is correct.

A thermally insulated closed vessel having $q = 0$, as it is mechanically stirred from outside.

$\therefore w > 0$ because work is done on the vessel.

Similarly, from First law of thermodynamics

$\Delta U = q + w$

As $q = 0$, so $\Delta U = w$

As $w > 0$, so $\Delta U > 0$

34. Option (4) is correct.

Formula used:

Bohr radius = $a_0 \times \frac{n^2}{z}$ n^{th} orbit

Where, n = No. of orbit

z = Atomic number

In the first excited state, e^- jumps from 1st orbit to 2nd orbit.

$\therefore n = 2$ and z = 2

By putting the value,

$$r^n = a_0 \times \frac{(2)^2}{2} = 2a_0$$

35. Option (4) is correct.

Statement I is correct.

$CH_3–O–CH_2–Cl$ will undergo S_N1 reaction though it is primary halide because CH_3-O-CH_2-Cl forms a carbocation during S_N1 which is resonance stabilised.

$$CH_3–O–CH_2–Cl \longrightarrow CH_3–O–\overset{\oplus}{C}H_2 + \overset{\ominus}{Cl}$$

$$CH_3–\ddot{O}–\overset{\oplus}{C}H_2 \longleftrightarrow CH_3–\overset{\oplus}{O}=CH_2$$

Statement II is correct.

$CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}–CH_2–Cl$ will not undergo S_N2 reaction very easily though it is a primary halide because due to steric hinderance neopentyl chloride is insert towards S_N2 reaction.

36. Option (1) is correct.

Statement I is correct. One mole of propyne reacts with excess of sodium to liberate half a mole of H_2 gas.

$$CH_3 – C \equiv CH + Na \rightarrow CH_3 – C \equiv C\bar{}\,\overset{+}{Na} + 1/2\, H_2\uparrow$$

Statement II is incorrect.

Four gram of propyne reacts with $NaNH_2$ to liberate NH_3, gas which occupies 2240 mL at STP.

Molecular weight of C_3H_4 (propyne) = 40 g

$$CH_3–C \equiv CH + NaNH_2 \rightarrow CH_3–C \equiv C\bar{}\,\overset{+}{Na} + NH_3$$

40 g 22,400 mL at STP

40 g propyne liberates 22,400 mL NH_3

So, 4 g propyne liberates x mL NH_3

$$\frac{40\,g}{4\,g} = \frac{22,400\,mL}{x}$$

$$x = \frac{22,400\,mL \times 4\,g}{40\,g} = 2240\ mL$$

37. Option (2) is correct.

Balanced reaction:

$CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$

Given pressure of CO_2 = 0.5 atm

Pressure of equilibrium = 0.8 atm

	$CO_{2(g)}$	+ $C_{(s)}$	$\rightleftharpoons$	$2CO_{(g)}$
I(atm)	0.5	–		–
C(atm)	$-x$	–		$+2x$
E(atm)	$0.5 - x$	–		$2x$

Total pressure at equilibrium

$= 0.5 - x + 2x = 0.8$ atm

$= 0.5 + x = 0.8$

$x = 0.3$ atm

$P_{CO_2} = 0.5 - x = 0.5 - 0.3 = 0.2$ atm

$P_{CO} = 2x = 2 \times 0.3 = 0.6$ atm

$$K_p = \frac{(P_{CO})^2}{P_{CO_2}} = \frac{0.6 \times 0.6}{0.2} = 1.8\,atm$$

38. Option (4) is correct.

$CH_3–\underset{\displaystyle COOH}{\overset{|}{CH}}–(CH_2)–\underset{\displaystyle COOCH_3}{\overset{|}{CH}}–CH_3$

According to the IUPAC rule, the maximum number of carbon atom in the main chain are six and the functional group of higher priority is carboxylic acid. So, the name ends with the suffix 'oic acid'.

$$CH_3–\overset{2}{\underset{{}_1COOH}{\overset{|}{CH}}}–\overset{3}{CH_2}–\overset{4}{CH_2}–\overset{5}{\underset{{}_6COOCH_3}{\overset{|}{CH}}}–CH_3$$

Correct IUPAC name is

6-methoxy carbonyl-2, 5-dimethylhexanoic acid.

39. Option (3) is correct.

During the electrolysis of concentrated sulphuric acid, the following reaction takes place:

At anode: $2H_2SO_4 \rightarrow H_2S_2O_8 + 2H^+ + 2e^-$

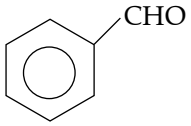
At cathode: $2e^- + 2H_2O \rightarrow H_2 + 2OH^-$

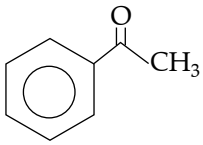
So, here at cathode, $S_2O_8^{2-}$ is formed.

which can be isolated in the form of $H_2S_2O_8$, i.e., Marshall acid.

40. Option (3) is correct.

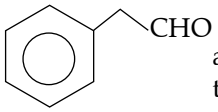
Aldehyde can be easily oxidises to carboxylic acid or carboxylate ion in the presence of mild or strong oxidising agent. Here, Fehling's reagent is a mild oxidising agent, which is used to distinguish aliphatic aldehyde from aromatic aldehyde as well as ketone. Fehling's test is mainly shown by aliphatic aldehyde, terminal α – hydroxyl carbonyl compounds, hemiacetal, or hemiketal etc.

(A)  aromatic aldehyde do not give this test.

(B)  ketone do not give this test.

(C) $HOCH_2–\overset{\overset{O}{\|}}{C}–(CHOCH)_3–CH_2OH$ terminal α hydroxyl carbonyl compound give this test.

(D) $CH_3–\overset{\overset{O}{\|}}{C}–H$ aliphatic aldehyde give this test.

(E)  aliphatic aldehyde give this test.

So, C, D and E can give this test.

41. Option (4) is correct.

Complex	Oxidation State of Fe	Strength of Ligand
$[Fe(NH_3)_6]Br_2$	+2	Strong
$[Fe(en)_3]Cl_3$	+3	Strong
$K_4[Fe(CN)_6]$	+2	Strong
$K_3[Fe(SCN)_6]$	+3	Weak

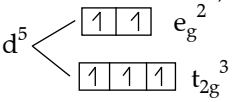
In $K_3[Fe(SCN)_6]$, the oxidation state of Fe is +3.

$\therefore$ its electronic configuration is:

Fe: $3d^6\ 4s^2$

Fe^{3+} : $3d^5\ 4s^0$

The value of CFSE, Δ_0 for d^5 configuration is:

d^5 

Δ_0 for octahedral: complex:

$(-0.4)\times nt_{2g}^{\ 3} + (+0.6) \times ne_g^{\ 2}$

$= (-0.4 \times 3) + (0.6 \times 2)$

$= 0$

42. Option (1) is correct.

We all know, elevation in boiling point depends on the number of solute particles present in the solution.

Here, urea is non-electrolyte while NaCl is an electrolyte, which dissociate to give two ions:

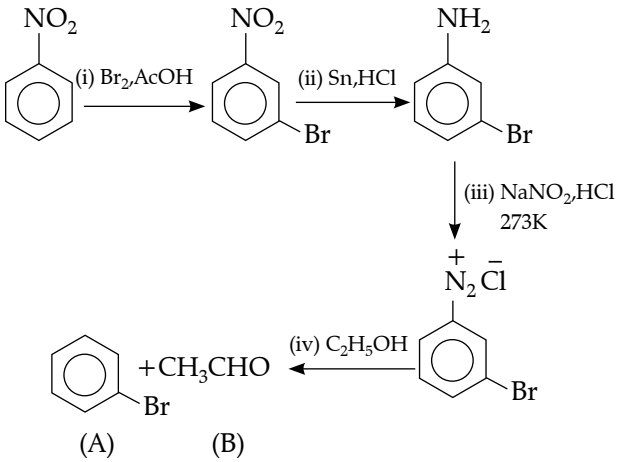
Let's assume M = m

	Solution	M	i	$\Delta T_b = ik_bm$
(i)	10^{-4} M NaCl	10^{-4}	2	$=2 \times k_b \times 10^{-4}$
(ii)	10^{-4} M Urea	10^{-4}	1	$=1 \times k_b \times 10^{-4}$
(iii)	10^{-3} M NaCl	10^{-3}	2	$=2 \times k_b \times 10^{-3}$
(iv)	10^{-2} M NaCl	10^{-2}	2	$=2 \times k_b \times 10^{-2}$

From above, it is clear that solution (iv) having highest ΔT_b value, so, the boiling point of solution (iv) is highest, while (ii) is the lowest.

So, order of increasing boiling point is follows: (ii) <(i) <(iii)<(iv)

43. Option (3) is correct.



(A) (B)

44. Option (2) is correct.

The oxidation state of Ni in $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ is +2, and 0 respectively.

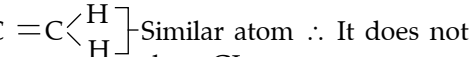
The hybridisation of the complex $[NiCl_4]^{2-}$ and $[Ni(CO_4)]$ is tetrahedral only. In both complexes, hybridisation on state of central metal (Ni) is sp^3.

$\therefore$ the geometry will be tetrahedral.

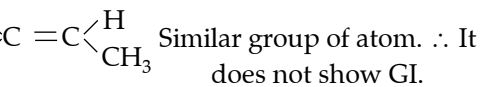
45. Option (1) is correct.

Following are the incorrect statements:

(A) Propene does not show geometrical isomerism because it has similar group attached to double bond.



(D) 2- methyl but -2- ene does not show geometrical isomerism because it has similar group attached to double bond.



(E) The melting point of trans is greater than that of cis because sterically hindered groups are present opposite to each other, which result effective packing among the trans molecule and hence, boiling point increases.

46. Correct answer is [45].

m. eq of $Ca(OH)_2$ left = m.eq. of HCl reacts

= 40 mL × 0.1 = 4 Meq. HCl.

M mol of $Ca(OH)_2$ reacts

$$= \frac{\text{Meq. of } Ca(OH)_2}{2} = \frac{4}{2} = 2\,\text{Mmol}$$

Similarly, half of the millimol of $Ca(OH)_2$ already reacts with CO_2

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

M mol of $Ca(OH)_2$ = M mol of CO_2

m mol of CO_2 = 2 M mol $= 2 \times 10^{-3}$ mol

from $PV = nRT$

$P = 1$ atm $T = 273$ K $n = 2 \times 10^{-3}$ mol

$$V = \frac{nRT}{P} = \frac{2\times10^{-3}\,\text{mol}\times0.0821\,L-\text{atm}\times273\,K}{1\,\text{atm}\;K-\text{mol}}$$

= 44.83 mL

= 45 mL = 45 cm^3

47. Correct answer is [20].

Formula used,

$$\text{Percentage of chlorine} = \frac{\text{atomic mass of Cl} \times \text{mass of AgCl} \times 100}{\text{molor mass of AgCl} \times \text{weight of organic compound}}$$

Given weight of organic compound = 180 mg

Mass of AgCl produced = 143.5 mg

Molar mass of AgCl = 143.5 g

$$\%\ \text{of Cl} = \frac{35.5\times143.5\text{mg}\times100}{143.5\text{gm}\times180\,\text{mg}} = 19.72$$

$\approx 20\%$

48. Correct answer is [6].

Molecule	Hybridisation	Geometry
SO_2	sp^2	Bent/Angular
$BeCl_2$	sp	Linear
CO_2	sp	Linear
N_3^-	sp	Linear
NO_2	sp^2	Bent/Angular
F_2O	sp^3	Bent/Angular
XeF_2	sp^3d	Linear
NO_2^+	sp	Linear
I_3^-	sp^3d	Linear
O_3	sp^3	Bent/Angular

49. Correct answer is [69].

$E_a = 191.48$ kJ mol^{-1} or 191480 J mol^{-1}

Frequency factor $A = 10^{20}$

R = 8.314 $JK^{-1}\,mol^{-1}$

$$\log_{10} K = \frac{-Ea}{2.303RT} + \log_{10} A$$

$$= \frac{-191480}{2.303\times8.314\times1000} + \log_{10} 10^{20}$$

$= -10.0004 + 20 \times 1$

$= -10.0004 + 20\log 10$

$\log_{10} K = 9.9995$

$K = \text{Antilog}_{10}\,9.9995$

$K = 9.9885 \times 10^9$

The required time for 50% of A to become B will be equal to $t_{\frac{1}{2}}$

$$t_{\frac{1}{2}} = \frac{0.693}{K}$$

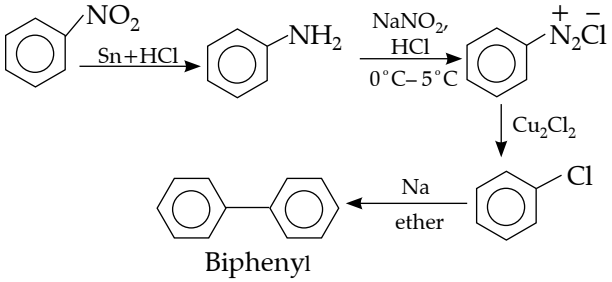
$$t_{\frac{1}{2}} = \frac{0.693}{9.9885\times10^9}$$

$$t_{\frac{1}{2}} = 0.06938\times10^{-9}\,\text{ps}$$

$= 0.06938 \times 10^{-9} \times 10^{12}$ ps

$t_{\frac{1}{2}} = 69.38$ ps

50. Correct answer is [154].



Biphenyl

The molar mass of biphenyl ($C_{12}H_{10}$) = 154 g mol^{-1}

Mathematics

Section A

51. Option (3) is correct.

A = {1, 2, 3} non-empty equivalent relation

$R_1 = \{(1, 1), (2, 2), (3, 3)\}$

$R_2 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$

$R_3 = \{(1, 1), (2, 2), (3, 3), (2,3), (3, 2)\}$

$R_4 = \{(1, 1), (2, 2), (3, 3), (1, 3), (3, 1)\}$

$R_5 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (1, 3), (3, 1), (2,3), (3, 2)\}$

$\therefore$ Number of relation $= 5$

52. Option (1) is correct.

$f(x + y) = f(x) f(y) \quad \forall \ x, y \in R$

From the functional equation $f(x + y) = f(x) f(y)$:

Put $y = 0, f(x) = f(x) = f(0)$

This means either $f'(0) = 1$ or $f(x) = 0$ for all x

Since we have $f'(0) = 4a > 0$, $f(x)$ cannot be identiacially zero Therefore $f(0) = 1$

This implies that f is an exponential function.

$f(x) = k^x$ for some constant k

$$f'(x) = k^x \ln k$$

$$f'(0) = \ln k = 4a$$

$$k = e^{4a}$$

$$f(x) = \left(e^{4a}\right)^x = e^{4ax}$$

$$f'(x) = 4ae^{4ax}$$

$$f''(x) = 16a^2 e^{4ax}$$

Now, $f''(x) - 3af'(x) - f(x) = 0$

$16a^2e^{4ax} - 12a^2e^{4ax} - e^{4ax} = 0$

$(4a^2 - 1)e^{4ax} = 0$

$4a^2 = 1$

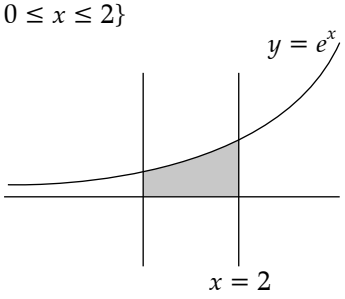
$a = \frac{1}{2}$ ($\because a > 0$)

$\therefore \quad y = e^{2x}$

$f(ax) = e^{4a(ax)} = e^{4a^2x} = e^x$

$\Rightarrow \quad f(ax) = e^x$

Area of region R = $\{(x, y): 0 \le y \le f(ax), 0 \le x \le 2\}$



$$a = \int_0^2 e^x dx$$

$$= \left[e^x\right]_0^2$$

$$= e^2 - 1$$

53. Option (4) is correct.

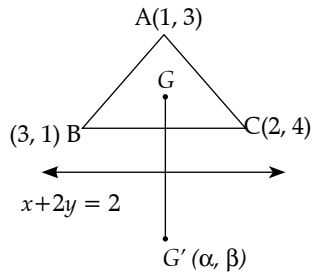
$\because$ ΔPQR is image of ΔABC with respect to line $x + 2y = 2$

Centroid of ABC

$G = \left(2, \frac{8}{3}\right)$

$\therefore$ Centroid of PQR is G′ (α, β)

That is image of G wrt $x + 2y - 2 = 0$



$$\frac{\alpha - 2}{1} = \frac{\beta - \frac{8}{3}}{2} = \frac{-2\left(2 + \frac{16}{3} - 2\right)}{1 + 4}$$

$$\Rightarrow \frac{\alpha - 2}{1} = \frac{\beta - \frac{8}{3}}{2} = \frac{-32}{15}$$

$$\alpha = 2 - \frac{32}{15} = \frac{-2}{15}$$

$$\beta = \frac{8}{3} - \frac{64}{15} = \frac{-24}{15}$$

$\therefore 15 (\alpha - \beta) = 22$

54. Option (4) is correct.

Given, z_1, z_2, z_3 are three complex numbers and lies on circle $|z| = 1$ and

$\text{Arg}(z_1) = \frac{-\pi}{4}$

$\text{Arg}(z_2) = 0$

$\text{Arg}(z_3) = \frac{\pi}{4}$

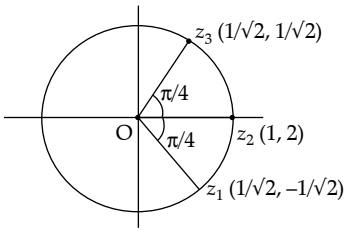
using euler's formula

$z = e^{-i\theta} = \cos(\theta) + i\sin(\theta)$

$\therefore \quad z_1 = \frac{1}{\sqrt{2}}(1 - i) \quad \bar{z}_1 = \frac{1}{\sqrt{2}}(1 + i)$

$z_2 = 1 + 0 \quad \bar{z}_2 = 1$

$z_3 = \frac{1}{\sqrt{2}}(1+i) \quad \bar{z}_3 = \frac{1}{\sqrt{2}}(1-i)$



$\therefore \ |z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1|^2$

$= \left|\frac{1}{\sqrt{2}}(1-i) + \frac{1}{\sqrt{2}}(1-i) + \frac{1}{2}(1+i)^2\right|^2$

$= \left|\sqrt{2} - \sqrt{2}i + \frac{1}{2}(2i)\right|^2$

$= \left|\sqrt{2} + \left(1-\sqrt{2}\right)i\right|^2$

$= 2 + (1-\sqrt{2})^2$

$= 2 + 1 + 2 - 2\sqrt{2}$

$= 5 - 2\sqrt{2}$

$\therefore \quad \alpha = 5, \beta = -2$

$\Rightarrow \alpha^2 + \beta^2 = 25 + 4 = 29$

55. Option (4) is correct.

$y = (\sec^{-1}x)^2 + (\text{cosec}^{-1}x)^2$

$= (\sec^{-1}x + \text{cosec}^{-1}x)^2 - 2\sec^{-1}x \ \text{cosec}^{-1}x$

$= \left(\frac{\pi}{2}\right)^2 - 2(\sec^{-1}x)\left(\frac{\pi}{2} - \sec^{-1}x\right)$

$= \frac{\pi^2}{4} + 2(\sec^{-1}x)^2 - \pi\sec^{-1}x$

$= 2\left((\sec^{-1}x)^2 - \frac{\pi}{2}\left(\sec^{-1}x\right) + \frac{\pi^2}{8}\right)$

$= 2\left(\left(\sec^{-1}x - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{8} - \frac{\pi^2}{16}\right)$

$= 2\left(\sec^{-1}x - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{8}$

$\therefore \ \sec^{-1}x \in [0,\pi] - \left\{\frac{\pi}{2}\right\}$

$\therefore$ minimum value of $y = 0 + \frac{\pi^2}{8}$

Maximum value of $y = 2\left(\frac{9\pi^2}{16}\right) + \frac{\pi^2}{8} = \frac{10\pi^2}{8}$

$\therefore$ Sum required $= 16\left(\frac{\pi^2}{8} + \frac{10\pi^2}{8}\right) = 22\pi^2$

56. Option (2) is correct.

x = No. of times a tail follows a head.

Outcome	x_i	p_i
HHH	0	$\frac{1}{8}$
HHT	1	$\frac{1}{8}$
HTH	1	$\frac{1}{8}$
THH	0	$\frac{1}{8}$
HTT	1	$\frac{1}{8}$
THT	1	$\frac{1}{8}$
TTH	0	$\frac{1}{8}$
TTT	0	$\frac{1}{8}$

$p(x=0) = \frac{4}{8} = \frac{1}{2}$

$p(x=1) = \frac{4}{8} = \frac{1}{2}$

$\mu = \sum p_i x_i = 0 + \frac{1}{2} = \frac{1}{2}$

$\sigma^2 = \sum p_i x_i^2 - \mu^2$

$= 0 + \frac{1}{2} - \frac{1}{4}$

$= \frac{1}{4}$

$\therefore \ 64(\mu + \sigma^2) = 64\left(\frac{3}{4}\right) = 48$

57. Option (3) is correct.

$a_1, a_2, \ldots$ GP

$a_1a_5 = 28$ and $a_2 + a_4 = 29$

Let a be the first term and 'r' be the common ratio

$a(ar^4) = 28$ and $ar + ar^3 = 29$

$a^2r^4 = 28$

$ar^2 = 2\sqrt{7}$

$\Rightarrow \frac{2\sqrt{7}}{r} + 2\sqrt{7}r = 29$

$\Rightarrow 2\sqrt{7}r^2 - 29r + 2\sqrt{7} = 0$

$\Rightarrow 2\sqrt{7}r^2 - 28r - r + 2\sqrt{7} = 0$

$\Rightarrow 2\sqrt{7}r(r-2\sqrt{7})-1(r-2\sqrt{7})=0$

$\Rightarrow (r-2\sqrt{7})(2\sqrt{7}r-1)=0$

$\Rightarrow r=2\sqrt{7},\ r=\dfrac{1}{2\sqrt{7}}$

$\therefore$ Increasing GP $\Rightarrow r=2\sqrt{7}$

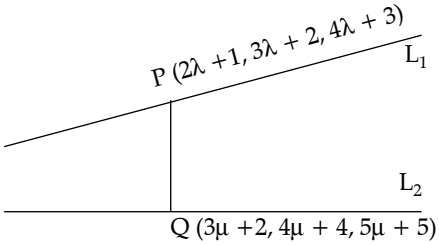
$\therefore \quad a=\dfrac{2}{2\sqrt{7}}$

$\therefore \quad a_6=ar^5=(2\sqrt{7})^4=(28)^2=784$

58. Option (4) is correct.

L_1: $\dfrac{x-1}{2}=\dfrac{y-2}{3}=\dfrac{z-3}{4}$

L_2: $\dfrac{x-2}{3}=\dfrac{y-4}{4}=\dfrac{z-5}{5}$



$\overrightarrow{PQ}=(3\mu-2\lambda+1)\hat{i}+(4\mu-3\lambda+2)\hat{j}+(5\mu-4\lambda+2)\hat{k}$

$\because\ \overrightarrow{PQ}\perp\vec{b}_1$ and $\overrightarrow{PQ}\perp\vec{b}_2$

$$\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&4\\3&4&5\end{vmatrix}=-\hat{i}+2\hat{j}-\hat{k}$$

$\overrightarrow{PQ}\ ||\ \vec{b}_1\times\vec{b}_2$

$\dfrac{3\mu-2\lambda+1}{-1}=\dfrac{4\mu-3\lambda+2}{2}=\dfrac{5\mu-4\lambda+2}{-1}$

$6\mu-4\lambda+2=-4\mu+3\lambda-2$

$-4\mu+3\lambda-2=10\mu-8\lambda+4$

$10\mu-7\lambda+4=0$ and $-14\mu+11\lambda-6=0$

$70\mu-49\lambda+28=0$

$-70\mu+55\lambda-30=0$

$6\lambda-2=0$

$\lambda=\dfrac{1}{3}$

$\mu=\left(\dfrac{7}{3}-4\right)\dfrac{1}{10}=\dfrac{-5}{3\times10}=\dfrac{-1}{6}$

$\therefore$ P $=\left(2\left(\dfrac{1}{3}\right)+1,\ 3\left(\dfrac{1}{3}\right)+2,\ 4\left(\dfrac{1}{3}\right)+3\right)$

$=\left(\dfrac{5}{3},\ 3,\ \dfrac{13}{3}\right)$

Equation of PQ

$\dfrac{x-\frac{5}{3}}{-1}=\dfrac{y-3}{2}=\dfrac{z-\frac{13}{3}}{-1}$

For $\left(\dfrac{14}{3},-3,\ \dfrac{22}{3}\right)$ satisfies the equation.

59. Option (1) is correct.

$e^{5(\ln x)^2+3}=x^8$

Taking ln both sides

$5(\ln x)^2+3=8\ln x$

$5(\ln x)^2-8\ln x+3=0$

$5(\ln x)^2-5\ln x-3\ln x+3=0$

$(5\ln x-3)(\ln x-1)=0$

$\ln x=\dfrac{3}{5},\ 1$

$x=e^{3/5},e^1$

Product $=e^{3/5}\times e^1=e^{8/5}$

60. Option (3) is correct.

$\sum_{r=1}^{n}T_r=\dfrac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$

$T_n=S_n-S_{n-1}=\dfrac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$

$=\dfrac{-(2n-3)(2n-1)(2n+1)(2n+3)}{64}$

$=\dfrac{(2n-1)(2n+1)(2n+3)(2n+5-2n+3)}{64}$

$=\dfrac{(2n-1)(2n+1)(2n+3)}{8}$

$\because \sum_{r=1}^{n}\dfrac{1}{T_r}=8\sum_{r=1}^{n}\dfrac{1}{(2n-1)(2n+1)(2n+3)}$

$=\dfrac{8}{4}\left(\sum_{r=1}^{n}\dfrac{1}{(2n-1)(2n+1)}-\dfrac{1}{(2n+1)(2n+3)}\right)$

$=2\left[\left(\dfrac{1}{1\cdot3}-\dfrac{1}{3\cdot5}\right)+\left(\dfrac{1}{3\cdot5}-\dfrac{1}{5\cdot7}\right)\cdots\right.$

$\left.\cdots\left(\dfrac{1}{(2n-1)(2n+1)}-\dfrac{1}{(2n+1)(2n+3)}\right)\right]$

$=2\left(\dfrac{1}{3}-\dfrac{1}{(2n+1)(2n+3)}\right)$

At $n\to\infty$ required value $=2\times\dfrac{1}{3}=\dfrac{2}{3}$

61. Option (4) is correct.

Total alphabets = 26

12 alphabets 13 alphabets

-------- M --------

Remaining alphabets = 25

Now, arrangement in alphabetical order

$= {}^{12}C_2 \times {}^{13}C_2 \times 1$

$= \frac{12\times11}{2\times1} \times \frac{13\times12}{2\times1}$

$= 5148$

62. Option (3) is correct.

$y^2dx + \left(x - \frac{1}{y}\right)dy = 0$

$y^2\frac{dx}{dy} + x = \frac{1}{y}$

$\frac{dx}{dy} + \left(\frac{1}{y^2}\right)x = \frac{1}{y^3}$

$P = \frac{1}{y^2}, Q = \frac{1}{y^3}$

I.F $= e^{\int Pdy} = e^{\int \frac{1}{y^2}dy} = e^{-\frac{1}{y}}$

Solution of DE

$x(\text{I}\cdot\text{F}) = \int Q\cdot(\text{IF})dy + c$

$xe^{-1/y} = \int \frac{1}{y^3}e^{-1/y}dy + C$

$e^{-1/y} = t \Rightarrow -1/y = \ln t$

$e^{-1/y}\frac{1}{y^2}dy = dt$

$xe^{-1/y} = \int -\ln t dt + C$

$xe^{-1/y} = -t(\ln t - 1) + C$

$xe^{-1/y} = -e^{-1/y}\left(\frac{-1}{y} - 1\right) + C$

$x = \frac{1}{y} + 1 + Ce^{1/y}$

$x(1) = 1 = 1 + 1 + Ce \qquad \Rightarrow C = \frac{-1}{e}$

$\therefore x\left(\frac{1}{2}\right) = 2 + 1 - \left(\frac{1}{e}\right)\cdot e^2 = 3 - e$

63. Option (2) is correct.

$y = x^2 + px - 3$

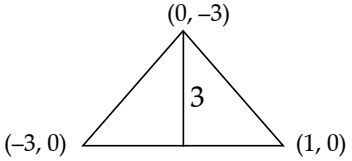
$\because x = 0 \quad y = -3$

$\therefore (0, -3)$ be point where parabola (cuts y axis)

$\therefore$ Equation of circle 'C'

$(x + 1)^2 + (y + 1)^2 = (-1 - 0)^2 + (-1 + 3)^2$

$\Rightarrow (x + 1)^2 + (y + 1)^2 = 5$



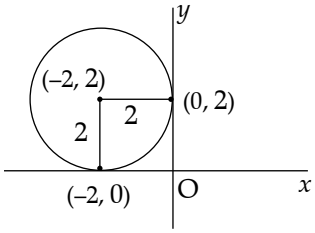
Circle cuts x-axis at $y = 0$

$\therefore (x+1)^2 = 4 \Rightarrow x + 1 = \pm 2 \Rightarrow x = 1, -3$

$\therefore$ Points are (1, 0), (−3, 0) and (0, −3)

Area of $\Delta = \frac{1}{2}\times 4 \times 3 = 6$

64. Option (1) is correct.



Equation of circle C: $(x + 2)^2 + (y - 2)^2 = 2^2$

Centre C_1 (−2, 2), $r_1 = 2$

$\because$ An another circle having centre C_2 (2, 5) and radius r intersects the circle C at exactly two points

$\therefore \quad |r_1 - r_2| < C_1C_2 < r_1 + r_2$

$|r - 2| < 5 < r + 2$

$-5 < r - 2 < 5$ and $r + 2 > 5$

$-3 < r < 7$ and $r > 3$

$\therefore \quad r \in (3, 7)$

$\alpha = 3 \quad \beta = 7$

$\therefore \quad 3\beta - 2\lambda = 21 - 6 = 15$

65. Option (3) is correct.

$f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$

$= 7\tan^6 x\left(\tan^2 x + 1\right) - 3\tan^2 x\left(\tan^2 x + 1\right)$

$= \sec^2 x\left(7\tan^6 x - 3\tan^2 x\right)$

$= \sec^2 x\tan^2 x\left(7\tan^4 x - 3\right)$

$I_1 = \int_0^{\frac{\pi}{4}} f(x)dx$

$= \int_0^{\frac{\pi}{4}} \sec^2 x\left(7\tan^6 x - 3\tan^2 x\right)dx$

Let, $\tan x = t \Rightarrow \sec^2 x\, dx = dt$

$= \int_0^1 \left(7t^6 - 3t^2\right)dt$

$= \left[t^7 - t^3\right]_0^1 = 0$

$I_2 = x\int_0^{\frac{\pi}{4}} f(x) - \int_0^{\frac{\pi}{4}} \left(1\cdot\int f(x)dx\right)dx$

$= 0 - \int_0^{\frac{\pi}{4}} (\tan^7 x - \tan^3 x)dx$

$I_2 = -\int_0^{\frac{\pi}{4}} \tan^3 x\left(\tan^4 x - 1\right)dx$

$= -\int_0^{\frac{\pi}{4}} \tan^3 x\left(\tan^2 x - 1\right)\left(\tan^2 x + 1\right)dx$

$$= -\int_0^{\frac{\pi}{4}} \tan^3 x\left(\tan^2 x - 1\right)\sec^2 x dx$$

$\tan x = t$

$\sec^2 x dx = dt$

$$= \int_0^1 t^3\left(1 - t^2\right)dt = \int_0^1 \left(t^3 - t^5\right)dt$$

$$= \left[\frac{t^4}{4} - \frac{t^6}{6}\right]_0^1$$

$$= \frac{1}{4} - \frac{1}{6}$$

$$= \frac{1}{12}$$

$\therefore 7I_1 + 12I_2 = 1$

66. Option (2) is correct.

Given, $f(x+y) = f(x)\cdot f'(y) + f'(x)\cdot f(y)$

Put $y = 0$

$f(x) = f(x)\cdot f'(0) + f'(x)f(0)$

$f(x) = f(x)\cdot f'(0) + f'(x)$ ....(i) $(\because f(0) = 1)$

Put $x = y = 0$

$f(0) = f'(0) + f'(0) \Rightarrow f'(0) = \frac{1}{2}$

Hence, equation (i) becomes

$$f(x) = \frac{f(x)}{2} + f'(x)$$

$$f'(x) = \frac{f(x)}{2}$$

$$\frac{f'(x)}{f(x)} = \frac{1}{2}$$

Integration both of sides

$$\ln f(x) = \frac{x}{2} + C$$

Put $x = 0 \Rightarrow C = 0$

$$\Rightarrow \ln f(x) = \frac{x}{2}$$

$$f(x) = e^{\frac{x}{2}}$$

$$\therefore f(n) = e^{\frac{n}{2}}$$

$$\log_e f(n) = \frac{n}{2}$$

$$\therefore \sum_{n=1}^{100} \log_e f(n) = \frac{1}{2}\sum_{n=1}^{100} n$$

$$= \frac{1}{2}\left(\frac{100}{2}(100+1)\right)$$

$$= 25 \times 101$$

$$= 2525$$

67. Option (1) is correct.

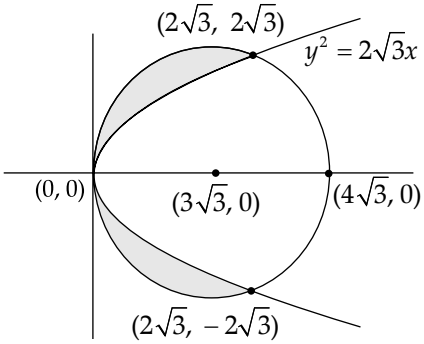
A = {1, 2, 3 ... 10}

$$B = \left\{\frac{m}{n} : m, n \in A, m < n \text{ and } \gcd(m,n) = 1\right\}$$

$m = 1 \Rightarrow n = 2, 3, 4 \ldots 10$	9 elements
$m = 2 \Rightarrow n = 3, 5, 7, 9$	4 elements
$m = 3 \Rightarrow n = 4, 5, 7, 8, 10$	5 elements
$m = 4 \Rightarrow n = 5, 7, 9$	3 elements
$m = 5 \Rightarrow n = 6, 7, 8, 9$	4 elements
$m = 6 \Rightarrow n = 7$	1 element
$m = 7 \Rightarrow n = 8, 9, 10$	3 elements
$m = 8 \Rightarrow n = 9$	1 element
$m = 9 \Rightarrow n = 10$	1 element

$\therefore n(B) = 31$

68. Option (3) is correct.



Circle: $\left(x - 2\sqrt{3}\right)^2 + y^2 = 12 \Rightarrow x = 2\sqrt{3} \pm \sqrt{12 - y^2}$

Parabola: $y^2 = 2\sqrt{3}x$

Point of intersection $\Rightarrow y^2 = 12 \Rightarrow y = \pm 2\sqrt{3}$

$\therefore x = +2\sqrt{3}$

Area of required region $= 2\int_0^{2\sqrt{3}} (x_1 - x_2)dy$

$$= 2\int_0^{2\sqrt{3}} \left(\frac{y^2}{2\sqrt{3}} - 2\sqrt{3} + \sqrt{12 - y^2}\right)dy$$

$$= 2\left[\frac{y^3}{3\sqrt{3}} - 2\sqrt{3}y + \frac{y}{2}\sqrt{12 - y^2} + \frac{12}{2}\sin^{-1}\frac{y}{2\sqrt{3}}\right]_0^{2\sqrt{3}}$$

$$2\left[4 - 12 + 0 + 6 \times \frac{\pi}{2}\right] = 6\pi - 16$$

69. Option (1) is correct.

E_1 = First selected ball is white.

E_2 = First selected ball is black.

A = Second selected ball is black.

$$P(E_2/A) = \frac{P(E_2)\cdot P(A/E_2)}{P(E_1)\cdot P(A/E_1) + P(E_2)P(A/E_2)}$$

$$= \frac{\frac{6}{10} \times \frac{5}{9}}{\frac{4}{10} \times \frac{6}{9} + \frac{6}{10} \times \frac{5}{9}}$$

$$= \frac{30}{24 + 30} = \frac{30}{54} = \frac{5}{9} = \frac{m}{n}$$

$\therefore m + n = 5 + 9 = 14.$

70. Option (3) is correct.

Foci of hyperbola (1, 14) (1, −12)

$\therefore$ $x = 1$ be the transverse axis

$|SP - S'P| = 2b$

$\Rightarrow 2b = |8 - 18|$

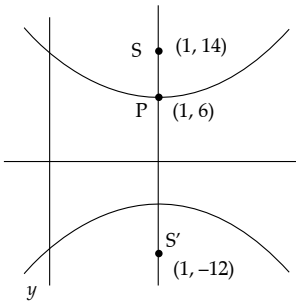
$\Rightarrow b = 5$

$SS' = 2be$

$26 = 2be$

$be = 13$

$e = \frac{13}{5}$



$\therefore$ Length of L R

$= \frac{2a^2}{b} = \frac{2b^2(e^2-1)}{b}$

$= 2(5)\frac{(144)}{25} = \frac{288}{5}$

Section B

71. Correct answer is [34].

$f(x) = \begin{cases} -3ax^2 - 2 & x < 1 \\ a^2 + bx & x \geq 1 \end{cases}$

$\because f(x)$ is differentiable.

$f'(x) = \begin{cases} -6ax & x < 1 \\ b & x > 1 \end{cases}$

$f(1^+) = f(1^-) = f(1)$

$\therefore -3a - 2 = a^2 + b$

$a^2 + 3a + b = -2$ ...(i)

$f'(1^+) = f'(1^-)$

$-6a = b$ ...(ii)

Solving (i) and (ii)

$a^2 - 3a + 2 = 0$

$(a-1)(a-2) = 0 \Rightarrow a = 1, 2$ $(\because a > 1)$

$\Rightarrow a = 2$

$b = -12$

Now, $y = f(x) = \begin{cases} -6x^2 - 2 & x < 1 \\ -12x + 4 & x \geq 1 \end{cases}$

$-6x^2 - 2 = -20$ $\quad -12x + 4 = -20$

$6x^2 = 18$ $\quad 12x = 24$

$x^2 = 3$ $\quad x = 2$

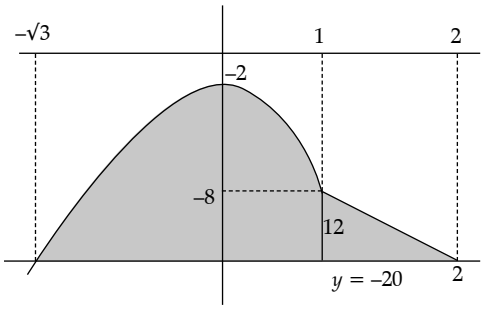
$x = \pm\sqrt{3}$

Since, we are in the domain $x < 1$, we have $x = -\sqrt{3}$.

$\therefore$ Required area =

$20(\sqrt{3}+1) - \left|\int_{-\sqrt{3}}^{1} ydx\right| + \frac{1}{2}\times 1\times 12$

$= 20(\sqrt{3}+1) - \left|\int_{-\sqrt{3}}^{1} -6x^2 - 2\right| + 6$



$= 20\sqrt{3} + 26 - \left|\left[2x^3 + 2x\right]_{-\sqrt{3}}^{1}\right|$

$= 20\sqrt{3} + 26 - \left|4 + 8\sqrt{3}\right|$

$= 20\sqrt{3} + 26 - 8\sqrt{3} - 4$

$= 22 + 12\sqrt{3}$

$\alpha = 22$

$\beta = 12$

$\alpha + \beta = 34$

72. Correct answer is [2035].

$\sum_{r=0}^{5} \frac{^{11}C_{2r+1}}{2r+2}$

$= \frac{1}{12}\sum_{r=0}^{5} {}^{12}C_{2r+2} \left(\frac{^{n+1}C_{r+1}}{n+1} = \frac{^{n}C_{r}}{r+1}\right)$

$= \frac{1}{12}\left({}^{12}C_2 + {}^{12}C_4 + {}^{12}C_6 + {}^{12}C_8 + {}^{12}C_{10} + {}^{12}C_{12}\right)$

$= \frac{1}{12}\left(2^{11} - 1\right)$

$= \frac{2^{11} - 1}{12}$

$= \frac{2047}{12} = \frac{m}{n}$

$\Rightarrow m - n = 2047 - 12 = 2035$

73. Correct answer is [34].

$|A| = -2$ Order of matrix $A = 3$

$\because \text{adj}(KA) = k^{n-1}\text{adj}A$

$\therefore |3adj(-6adj3A)|$

$|3adj(-6\times 3^{3-1} adjA)|$

$= |3(-6\times 3^2)^{3-1} adj\, adjA|$

$= |3\times 6^2 \times 3^4 adj\, adjA|$

$= 3^3 \times (6^2)^3 (3^4)^3 |(adj\, adj\, A)|$ $\quad \because |kA| = k^n(A)$

$= 3^3 \times 2^6 \times 3^6 \times 3^{12}$ $\quad |A|^{(n-1)^2}$

$= 2^6 \times 3^{21} \times (-2)^4$

$= 2^{10} \times 3^{21}$

$\therefore\ m + n = 10$

$mn = 21$ and $m > n$

$\Rightarrow m = 7$ and $n = 3$

$4m + 2n = 28 + 6 = 34$

74. Correct answer is [216].

$$L_1 : \frac{x-1}{3} = \frac{y-1}{-1} = \frac{z+1}{0} = \lambda \Rightarrow P(3\lambda+1, -\lambda+1, -1)$$

$$L_2 : \frac{x-2}{2} = \frac{y}{0} = \frac{z+4}{\alpha} = \mu \Rightarrow Q(2\mu+2, 0, \alpha\mu-4)$$

$\because$ L_1 and L_2 intersects

$$\therefore 3\lambda+1 = 2\mu+2 \quad \Big| \quad -\lambda+1 = 0 \quad \Big| \quad -1 = \alpha\mu - 4$$

$$\lambda = 1$$

$$4 = 2\mu + 2 \qquad\qquad \Rightarrow -1 = \alpha - 4$$

$$\mu = 1 \qquad\qquad \alpha = 3$$

B = (4, 0, −1)

$$\therefore L_2 : \frac{x-2}{2} = \frac{y}{0} = \frac{z+4}{3}$$

$$\overrightarrow{AM} \perp \left(2\hat{i} + 0j + 3\hat{k}\right)$$

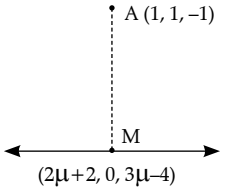
$$\Rightarrow \overrightarrow{AM} \cdot \left(2\hat{i} + 3\hat{k}\right) = 0$$

$$\Rightarrow 2(2\mu+1) + 3(3\mu-3) = 0$$

$$\Rightarrow\ 4\mu + 2 + 9\mu - 9 = 0$$

$$\Rightarrow 13\mu = 7$$

$$\Rightarrow\ \mu = \frac{7}{13}$$



$$\therefore 26\alpha(PB)^2$$

$$\Rightarrow 26(3)\frac{36}{13}$$

$$\Rightarrow 6 \times 36 = 216$$

75. Correct answer is [16].

$\vec{b} = \lambda\hat{i} + 4\hat{k} \quad \lambda > 0$

$\vec{c}$ = Projection of $\vec{b}$ on vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$

$$\left(\left|\vec{b}\cdot\vec{a}\right|\cos\theta\right)\hat{a}$$

$$= \left(\frac{\vec{b}\cdot\vec{a}}{|a|^2}\right)\hat{a}$$

$$\Rightarrow \frac{(\lambda+8)}{9} \times \left(\hat{i} + 2\hat{j} + 2\hat{k}\right)$$

$$|\vec{a} + \vec{c}| = 7$$

$\because |\vec{a}|^2 + |\vec{c}|^2 + 2\,\vec{a}\cdot\vec{c} = 49$

$$\Rightarrow 9 + \left(\frac{\lambda+8}{9}\right)^2 (9) + 2\left(\frac{\lambda+8}{9}\right)9 = 49$$

$$\Rightarrow 9 + \frac{(\lambda+8)^2}{9} + 2(\lambda+8) = 49$$

Let $\lambda + 8 = t$

$$t^2 + 81 + 18t = 49 \times 9$$

$$t^2 + 18t - 360 = 0$$

$$t^2 + 30t - 12t - 360 = 0$$

$$(t+30)(t-12) = 0$$

$$t = -30, 12$$

$$\lambda = -38, 4$$

$$\lambda > 0\ ,\ \lambda = 4$$

$$\vec{c} = \frac{12}{9}\left(\hat{i} + 2\hat{j} + 2\hat{k}\right)$$

$$= \frac{4}{3}\left(\hat{i} + 2\hat{j} + 2\hat{k}\right)$$

So, area of parallelgram

$$= b \times c = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 0 & 4 \\ \frac{4}{3} & \frac{8}{3} & \frac{8}{3} \end{vmatrix}$$

Expanding along the first row:

$$= \hat{i}\begin{vmatrix} 0 & 4 \\ \frac{8}{3} & \frac{8}{3} \end{vmatrix} - \hat{j}\begin{vmatrix} 4 & 4 \\ \frac{4}{3} & \frac{8}{3} \end{vmatrix} + \hat{k}\begin{vmatrix} 4 & 0 \\ \frac{4}{3} & \frac{8}{3} \end{vmatrix}$$

$$= \hat{i}\left(0\frac{8}{3} - 4\frac{8}{3}\right) - \hat{j}\left(4\frac{8}{3} - 4\frac{4}{3}\right) + \hat{k}\left(4\frac{8}{3} - 0\frac{4}{3}\right)$$

$$= \hat{i}\left(-\frac{32}{3}\right) - \hat{j}\left(\frac{32}{3} - \frac{16}{3}\right) + \hat{k}\left(\frac{32}{3}\right)$$

$$= -\frac{32}{3}\hat{i} - \frac{16}{3}\hat{j} + \frac{32}{3}\hat{k}$$

Magnitude:

$$\sqrt{\left(-\frac{32}{3}\right)^2 + \left(-\frac{16}{3}\right)^2 + \left(\frac{32}{3}\right)^2}$$

$$\sqrt{\frac{1024}{9} + \frac{256}{9} + \frac{1024}{9}}$$

$$\sqrt{\frac{2304}{9}} = \sqrt{256} = 16$$

■■■